RENTAL MARKETPLACE PRODUCT ANALYTICS

Executive Insights & Recommendations

Five analyses · synthetic product data + Realtor.com-schema market data
Prepared from the warehouse marts described in the project README

Executive Summary

This report synthesizes five analyses run against the Rental Marketplace Product Analytics Platform: a funnel drop-off analysis, a cohort retention study, an A/B test readout, a revenue forecast, and a market opportunity scoring model. Five concrete, data-backed recommendations follow, along with the two real data-quality issues found and fixed during the build.

1 Fix mobile onboarding before scaling mobile paid-social spend

Mobile-acquired landlords from paid social abandon between "property added" and "listing published" at a `40.9%` rate, versus `1.5%` for referral-acquired desktop landlords. Paid social is also the highest-CAC channel, meaning acquisition spend is disproportionately producing signups that never reach a published listing.

> **Action:** Simplify the mobile property-setup flow (fewer required fields, save-and-resume) before increasing mobile paid-social budget. Re-measure abandonment by device after the fix ships.

2 Make "enable rent collection" the activation goal

Landlords whose first adopted feature is online rent collection retain at `80.1%` by month 6, versus `72.4%` for landlords who only use listing tools first. The effect holds after accounting for portfolio size and acquisition channel.

> **Action:** Update onboarding messaging and in-product nudges to point newly-activated landlords toward enabling rent collection in their first session, not just toward publishing a listing.

3 Ship the guided onboarding checklist

The Guided Property Onboarding A/B test produced a `+6.29 percentage point` absolute lift in 7-day activation (18.1% relative), 95% CI [+3.3%, +9.3%], $p<0.0001$. The sample-ratio-mismatch check passed. Thirty-day paid conversion also moved directionally positive.

> **Action:** Ship to 100% of new landlords. Follow with a dedicated mobile-onboarding experiment, since finding #1 shows that's where the largest remaining opportunity sits.

4 Plan around the Holt-trend forecast, not the optimistic case

A Holt linear-trend model backtested at `2.9% MAPE` against the last 6 months of actuals, dramatically outperforming a seasonal-naive baseline (`39.6% MAPE`).

> **Action:** Use the Holt forecast as the base case for budget and hiring planning. Treat the guided-onboarding rollout's activation lift as directional upside on top of that base case, not yet reflected in the forecast.

5 Prioritize Phoenix, Orlando, Austin, and Atlanta for expansion

These four metros rank in the top 5 by opportunity score under all three tested weighting schemes (demand-heavy, whitespace-heavy, and the balanced base case), meaning the recommendation is robust to how much the business trusts "demand" vs. "whitespace" vs. "current platform performance."

Metro	Demand score	Platform landlords	Activation	Paid conv.	Opportunity score
Phoenix, AZ	87.3	2,517	41.3%	11.1%	0.678
Orlando, FL	83.3	2,656	42.0%	11.5%	0.654
Austin, TX	84.5	3,038	42.5%	11.2%	0.643
Atlanta, GA	83.1	2,524	40.1%	10.7%	0.637

> **Action:** Prioritize these four for sales/marketing investment. Treat markets that only rank highly under a single weighting (e.g. San Antonio under whitespace-heavy) as lower-confidence picks pending further market-specific due diligence.

A Data-Quality Note

Two real bugs were found and fixed while building this platform, rather than hidden or worked around:

1. **Funnel analysis initially showed near-zero listing-publish abandonment** (<1%) across every segment — not a real finding, but a gap in the synthetic-data generator, which originally published every added property automatically. Device/channel-driven publish propensity was added and the full pipeline regenerated; finding #1 above is from the corrected data.
2. **The revenue-reconciliation quality test initially failed by 1,311 landlords** (mart said 3,102 active paid landlords; source staging said 4,413). Root cause: the MRR mart's subscription-month expansion only unrolled 24 months, truncating anyone subscribed longer than 2 years on a 4-year-old platform. A separate bug in the pipeline runner's idempotency check meant fixes weren't even taking effect on reruns. Both are fixed; the check now reconciles to a diff of 16 (0.4%, explained by month-boundary timing).

Both are documented on the dashboard's Data Quality page rather than scrubbed from the project history — catching exactly this kind of thing before a stakeholder sees a wrong number is the entire point of a quality-test layer.
